MODULE 19 · SECTION 3 - STORING THE DATA

# Choosing a Storage Service

The decision by access pattern, and the one phrase in each question that names the store.

SECTION 3 of 5 EXAM WEIGHT 20% MODULE 19 of 45

admin@sthithapragna.com

Choose storage by **how the data will be read**, not by how big it is.

# Eight access patterns, eight stores

| ACCESS PATTERN | STORE | THE PHRASE THAT DECIDES |
|---|---|---|
| Regional transactions | Cloud SQL | MySQL, PostgreSQL or SQL Server, lift and shift |
| Transactions plus analytics | AlloyDB | PostgreSQL-compatible, analytics on live data |
| Global transactions | Spanner | Global, strongly consistent, 99.999% |
| Wide-column, by key | Bigtable | Low-latency reads and writes at huge scale |
| Documents for apps | Firestore | Mobile, offline sync, flexible documents |
| Cache | Memorystore | Sub-millisecond, in front of a database |
| Objects and files | Cloud Storage | Files, images, archives, the data lake |
| Analytics | BigQuery | Ad hoc SQL over terabytes |

THE PATTERN

Find the phrase first. The product name usually follows from it.

# Where Larkspur keeps what

| DATA | STORE | WHY |
|---|---|---|
| Live reading per meter, for the app | Bigtable | Millions of keys, millisecond lookups |
| Tariff service | Cloud SQL for PostgreSQL | A standard relational app |
| Hot tariff lookups | Memorystore | Sub-millisecond reads in front of Cloud SQL |
| Warehouse and history analytics | BigQuery | SQL over billions of readings |
| Raw files, manuals, archives | Cloud Storage | Cheap, durable objects |

THE POINT

One company, five stores. Real systems mix them, and the exam expects you to.

# Cloud SQL, AlloyDB or Spanner?

| | CLOUD SQL | ALLOYDB | SPANNER |
|---|---|---|---|
| Engines | MySQL, PostgreSQL, SQL Server | PostgreSQL | GoogleSQL or PostgreSQL dialect |
| Scale | One primary, read pools | One primary, read pools | Horizontal, virtually unlimited |
| Best SLA | 99.99%, Enterprise Plus | Zonal failover built in | Up to 99.999%, multi-region |
| Analytics | Use a replica or BigQuery | Columnar engine, on live data | Columnar engine, Data Boost |
| Signal | Lift and shift | PostgreSQL plus analytics | Global, strongly consistent |

THE TIEBREAKER

Writes from several regions at once, with no conflicts, is the one job only Spanner does.

# Spanner editions, briefly

- **Standard**: regional only, 99.99%. Relational and key-value.
- **Enterprise**: adds Spanner Graph, full-text and vector search, the columnar engine, autoscaling.
- **Enterprise Plus**: dual- and multi-region, up to **99.999%**, geo-partitioning.
- Capacity starts at **100 processing units**. One node is 1,000.

> THE SIGNAL
>
> 'Multi-region' or 'five nines' in a Spanner question means Enterprise Plus.

# Bigtable: a giant sorted map

- **Billions of rows**, thousands of columns, petabytes. One row key, sorted.
- Google quotes **low single-digit millisecond** latency, and up to 17K requests a second per node.
- Atomic on **one row only**. Not relational, and not for ad hoc analytics.
- Add clusters to replicate. Multi-region instances reach **99.999%**.

> THE SIGNAL
>
> Time series, IoT, ad tech and fraud scoring are Bigtable's home ground.

# Firestore: documents for apps

- A **serverless document database**. No sharding, no maintenance windows.
- Client SDKs give **real-time updates** and **offline sync** for mobile and web.
- ACID multi-document transactions, multi-region, up to **99.999%**.
- **Enterprise edition** adds **MongoDB compatibility**. Datastore apps keep working.

> THE SIGNAL
>
> 'Mobile app, works offline, syncs when back online' points straight to Firestore.

# Memorystore: the fast lane

- Managed **Valkey**, **Redis** and **Memcached**, with **sub-millisecond** reads.
- Sits **in front of** a database, holding the answers asked for most.
- Valkey: cluster mode, up to five replicas per shard, across zones.
- A cache, not the **system of record**. Plan for it to be empty.

> THE SIGNAL
>
> 'Sub-millisecond' plus 'the same few items read constantly' means a cache.

# Cloud Storage or BigQuery?

CLOUD STORAGE

## Objects of any size

- Files, images, PDFs, backups
- The landing zone and the lake
- Classes and lifecycle: module 22
- Read by every engine

BIGQUERY

## Analytics at any scale

- Ad hoc SQL over petabytes
- Serverless, columnar storage
- Not for millisecond lookups
- Not for heavy row-by-row updates

THE TRAP

BigQuery answers questions about data. It is not the database behind an app's every click.

# Six choices that look right and are not

| THE TEMPTING CHOICE | WHY IT FAILS |
| --- | --- |
| **BigQuery** behind a mobile app | Built for scans, not millisecond lookups |
| **Bigtable** for joins and transactions | One row atomic, no relational joins |
| **Cloud SQL** for writes in three regions | One primary, in one region |
| **Spanner** for a small regional app | Works, but costs more than it needs to |
| **Memorystore** as the only copy | A cache can be emptied |
| **Firestore** for warehouse analytics | Built for app reads, not big scans |

THE PATTERN

Most wrong answers use a real product for the wrong access pattern.

> The customer app shows each of 3.1 million meters' latest reading. Writes arrive at up to 10,000 a second in the evening.
>
> Every app screen reads one meter by its ID, and must respond in a few milliseconds. A Cloud SQL prototype could not keep up.

## Which store should hold the live readings?

A Cloud SQL for PostgreSQL, on a larger machine with an index

B A BigQuery table, queried by the app for each screen it shows

C Bigtable, with a row for each meter, keyed by its meter ID

D Memorystore as the only store, with no database behind it

◆ PAUSE AND DECIDE

## Which store should hold the live readings?

A. A bigger Cloud SQL
*One primary still takes every write. The prototype already hit that wall.*

B. BigQuery per screen
*Built for scans, not millisecond lookups for every screen.*

C. Bigtable by meter
Key lookups in milliseconds, and writes that scale across nodes.

D. Memorystore alone
*Fast, but a cache is not the system of record.*

THE SIGNAL

Millions of keys, heavy writes, single-key reads in milliseconds: Bigtable.

> A ticketing company sells seats for concerts in Europe, America and Asia from servers in all three regions at once.
>
> A seat must never be sold twice, even during a regional outage, and the business wants five nines of availability.

## Which database fits this best?

A Spanner, multi-region, in the Enterprise Plus edition

B Cloud SQL, with read replicas in each of the three regions

C Bigtable, with a cluster in each region and multi-cluster routing

D AlloyDB, with a read pool of nodes added in each region

◆ PAUSE AND DECIDE

## Which database fits this best?

A **Spanner Enterprise Plus**
Writes everywhere, strongly consistent, up to 99.999%.

B **Cloud SQL replicas**
*Replicas only read. Every sale still goes to one region.*

C **Bigtable multi-cluster**
*Eventually consistent, so two regions could sell one seat.*

D **AlloyDB read pools**
*Read pools only read. Writes still go to one primary.*

THE SIGNAL

Writes in several regions, never conflicting, five nines: Spanner Enterprise Plus, multi-region.

> Larkspur's tariff team wants to keep using PostgreSQL. It also wants pricing analysts to run heavy analytical queries on live tariff and contract data.
>
> Those queries must not slow down transactions, and the team does not want a separate copy in BigQuery.

## Which service fits best?

A Cloud SQL for PostgreSQL, with one large read replica for analysts

B Spanner, using its PostgreSQL dialect for the tariff application

C A nightly export of the tariff tables into BigQuery for analysts

D AlloyDB for PostgreSQL, with its columnar engine turned on

◆ PAUSE AND DECIDE

## Which service fits best?

A **Cloud SQL replica**

*Offloads the reads, but row storage makes heavy analytics slow.*

B **Spanner PostgreSQL**

*A bigger move than needed, for a regional workload.*

C **Nightly BigQuery export**

*A separate copy, a day old, which the team ruled out.*

D **AlloyDB columnar**

PostgreSQL, with fast analytics on live data and no second copy.

THE SIGNAL

'PostgreSQL' plus 'analytics on live data' plus 'no second copy': AlloyDB.

> Larkspur's pricing engine reads the same few hundred active tariffs about 5,000 times a second. Cloud SQL's CPU is close to its limit.
>
> Each lookup should take well under a millisecond. Tariffs change only a few times a day.

## What should the team add?

A. A larger Cloud SQL machine, sized for the peak read load

B. A copy of the tariff table in Bigtable, for the pricing engine

C. Memorystore for Valkey, as a cache in front of Cloud SQL

D. BigQuery BI Engine, holding the tariffs in memory for queries

◆ PAUSE AND DECIDE

# What should the team add?

A. A larger machine

*Pays to read the same rows thousands of times, still not sub-millisecond.*

B. A Bigtable copy

*Milliseconds, not sub-millisecond, and a second store to sync.*

C. Memorystore cache

Sub-millisecond reads of hot data, and Cloud SQL's load falls.

D. BI Engine

*Accelerates BigQuery dashboards, not an application's lookups.*

THE SIGNAL

Same few items, read constantly, sub-millisecond: a cache in front of the database.

> Larkspur's field engineers need a tablet app for job sheets. Many meters are in basements with no signal.
>
> Engineers must keep working offline, and their changes must sync automatically when signal returns. Job sheets vary in shape.

## Which store should back the app?

A Cloud SQL, reached through an API that the tablets call

B Firestore, with its client SDKs and offline support enabled

C Bigtable, with a row for each job sheet, keyed by job ID

D Cloud Storage, with a separate JSON file saved for each job sheet

◆ PAUSE AND DECIDE

# Which store should back the app?

A Cloud SQL via an API

*No signal, no API. The app stops working in the basement.*

B Firestore with offline

Works offline, syncs on reconnect, and fits varied documents.

C Bigtable

*Server-side only, with no offline client sync.*

D JSON files

*No offline sync, no queries, and conflicting edits to manage.*

THE SIGNAL

Offline mobile work with automatic sync and flexible documents: Firestore.

> Larkspur wants to keep 18,000 pages of engineers' manuals as PDFs, plus site photos, for the field assistant and for archive.
>
> Some files are hundreds of megabytes. They are written once and read occasionally.

## Where should the files live?

A In a BigQuery table, with each file stored in a BYTES column

B In Bigtable, with each file split across the cells of one row

C In Firestore, with one document for each manual or photo

D In a Cloud Storage bucket, as objects, read where they sit

◆ PAUSE AND DECIDE

## Where should the files live?

A BigQuery BYTES

*Analytics storage, not a file store for large binaries.*

B Bigtable cells

*Cells are for small values. Large files fight the design.*

C Firestore documents

*Documents are limited to 1 MiB, far below these files.*

D Cloud Storage objects

Built for files of any size, cheap, and readable by every engine.

THE SIGNAL

Files of any size, written once, read occasionally: Cloud Storage.

> Analysts want to explore eight years of Larkspur's readings, about 1.2 PB, with ad hoc SQL.
>
> Queries join readings to customers and tariffs, and scan billions of rows. Nobody wants to manage servers.

## Which store fits this best?

A. Bigtable, using its SQL support for the analysts' questions

B. Cloud SQL for PostgreSQL, on its largest available machine

C. BigQuery, with the readings in partitioned, clustered tables

D. Spanner Enterprise, with its columnar engine switched on for analysts

◆ PAUSE AND DECIDE

## Which store fits this best?

A Bigtable SQL

*Great for key lookups, but ad hoc joins at this scale are not its job.*

B Largest Cloud SQL

*A single machine cannot scan a petabyte quickly.*

C BigQuery

Serverless, columnar, built for ad hoc joins over petabytes.

D Spanner columnar

*Analytics on operational data, not an eight-year warehouse.*

THE SIGNAL

Ad hoc SQL, joins, petabytes, no servers: BigQuery.

> A payroll application runs on MySQL 8.0, with a 200 GB database and modest load in one region.
>
> The company wants to move it to Google Cloud with no code changes, and a 99.99% availability SLA that includes maintenance.

## Which service fits this best?

A Spanner, after rewriting the application's queries for GoogleSQL

B AlloyDB, migrated with Database Migration Service

C Bigtable, with each payroll record stored as a row

D Cloud SQL for MySQL, in the Enterprise Plus edition

◆ PAUSE AND DECIDE

## Which service fits this best?

A **Spanner**

*A rewrite, which the company ruled out, for a modest regional load.*

B **AlloyDB**

*PostgreSQL only. A MySQL app would need changes.*

C **Bigtable**

*Not relational, so payroll queries would need rewriting.*

D **Cloud SQL Enterprise Plus**

MySQL as is, and 99.99% including maintenance.

THE SIGNAL

MySQL, no code changes, regional, 99.99% with maintenance: Cloud SQL Enterprise Plus.

# Module 19 in five lines

- Choose by **access pattern**: by key, transaction, document, cache, file or scan.
- **Cloud SQL** lift and shift. **AlloyDB** PostgreSQL plus analytics. **Spanner** global writes.
- **Bigtable** for huge key-value scale in milliseconds. **Firestore** for app documents and offline sync.
- **Memorystore** is a cache. **Cloud Storage** holds files. **BigQuery** answers analytics.
- Most wrong answers use a real product for the **wrong access pattern**.

COMING UP

Next: module 20, Bigtable row keys and performance.